

4	1-10	2	EN / ZH-CN
connected research modules	portfolio holdings	valuation methods	localized interface

THE PROBLEM

Early-career analysts often split filings, valuation, portfolio risk, and written conclusions across unrelated tools. Dates and return bases drift, unsupported data becomes zero, and a single fair-value number can hide the assumptions that created it.

THE PRODUCT

InvestIQ joins official filing facts, explicit calculation contracts, scenario valuation, portfolio impact, and an evidence-linked memo. The primary audience is finance students, aspiring equity-research analysts, and asset-management interns; plain-language explanations keep the same workflow usable for self-directed investors.

CORE WORKFLOW

1. Search a US ticker and inspect latest-reported SEC annual facts with filing receipts.
2. Review five-year growth, margins, cash flow, and explicit data gaps.
3. Run editable FCFF DCF, bull/base/bear ranges, sensitivity tables, and sourced peer multiples.
4. Diagnose portfolio return, downside risk, benchmark relationship, attribution, correlation, and concentration.
5. Preserve evidence and analyst-owned interpretation in a printable one-page memo.

MODEL GOVERNANCE

- Ranges before point estimates.
- Reported facts separated from user assumptions.
- Unavailable remains unavailable - never silent zero.
- Latest-reported facts never backfilled into historical analysis.
- VaR is a historical loss threshold, not a loss ceiling.
- Future DCA estimates are disabled.

TECHNICAL STACK

Next.js, TypeScript, React, CSS Modules, Recharts, Vitest, SEC EDGAR APIs, same-origin server routes, Git/GitHub, and Vercel-ready deployment. Domain math is implemented as pure functions with hand-calculated and boundary-value tests.

SELECT ANALYTICS

Revenue and EPS growth • operating and net margin • constructed free cash flow • FCFF DCF • EV-to-equity bridge • WACC x terminal-growth sensitivity • comparable multiples • CAGR • volatility • Sharpe • Sortino • beta • Jensen-style alpha • maximum drawdown • historical one-day VaR • correlation • HHI and effective holdings

EVIDENCE BASE

SEC EDGAR API and developer guidance • CFA Institute material on Sortino ratio, market-risk measurement, and model communication • Damodaran firm-valuation and terminal-value framework

PORTFOLIO DELIVERABLES

Public web app • source repository • methodology • project report • AAPL research memo sample • three-minute demo script • one-page summary